TD04 - Branches + Merge

Part 2 - Inspect the current branch

```
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (BrancheSakti) $ git branch
* BrancheSakti
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (BrancheSakti) $ git status
On branch BrancheSakti
Your branch is ahead of 'origin/BrancheSakti' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

Part 3 - Create a feature branch

```
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (BrancheSakti) $ git switch -c feature/display-observatio
n-dates origin/main
branch 'feature/display-observation-dates' set up to track 'origin/main'.
Switched to a new branch 'feature/display-observation-dates'
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (feature/display-observation-dates) $ git branch
  BrancheSakti
* feature/display-observation-dates
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (feature/display-observation-dates) $ git status
On branch feature/display-observation-dates
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (feature/display-observation-dates) $ git log --oneline -
3
c388ccd (HEAD -> feature/display-observation-dates, origin/main, origin/HEAD) Add files via upload
5207db8 first td session
ba5c9f4 Update Readme
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (feature/display-observation-dates) $
```

Part 4 - Implement the small feature

```
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (feature/display-observation-dates) $ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Observations : 21
First date   : 2026-09-01
Last date    : 2026-09-30
First close  : 250.00 USD
Last close   : 266.20 USD

Benchmark
SP500 - S&P 500
Observations : 21
First date   : 2026-09-01
Last date    : 2026-09-30
First close  : 6600.00
Last close   : 6742.00

Period: 1 month
Interval: Daily
```

Part 5 - Inspect and commit the branch change

```
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (feature/display-observation-dates) $ git status
On branch feature/display-observation-dates
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   src/main.py

no changes added to commit (use "git add" and/or "git commit -a")
```

```
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (feature/display-observation-dates) $ git diff
diff --git a/src/main.py b/src/main.py
index 884b045..1b342e5 100644
--- a/src/main.py
+++ b/src/main.py
@@ -25,6 +25,12 @@ def load_prices():
 def filter_prices(prices, ticker):
     return [row for row in prices if row["ticker"] == ticker]

+def get_first_date(prices):
+    return prices[0]["date"]
+
+
+def get_last_date(prices):
+    return prices[-1]["date"]

 def main():
     instruments = load_instruments()
@@ -39,22 +45,34 @@ def main():
     instrument_latest = instrument_prices[-1]
     benchmark_latest = benchmark_prices[-1]

+    instrument_first_date = get_first_date(instrument_prices)
+    instrument_last_date = get_last_date(instrument_prices)
+
+    benchmark_first_date = get_first_date(benchmark_prices)
+    benchmark_last_date = get_last_date(benchmark_prices)
+
```

```
@SaktiJaga ➜/workspaces/esilv-marketpulse-g07-totally_spies (feature/display-observation-dates) $ git add src/main.py
@SaktiJaga ➜/workspaces/esilv-marketpulse-g07-totally_spies (feature/display-observation-dates) $ git diff --staged
diff --git a/src/main.py b/src/main.py
index 884b045..1b342e5 100644
--- a/src/main.py
+++ b/src/main.py
@@ -25,6 +25,12 @@ def load_prices():
 def filter_prices(prices, ticker):
     return [row for row in prices if row["ticker"] == ticker]

+def get_first_date(prices):
+    return prices[0]["date"]
+
+
+def get_last_date(prices):
+    return prices[-1]["date"]

 def main():
     instruments = load_instruments()
@@ -39,22 +45,34 @@ def main():
     instrument_latest = instrument_prices[-1]
     benchmark_latest = benchmark_prices[-1]

+    instrument_first_date = get_first_date(instrument_prices)
+    instrument_last_date = get_last_date(instrument_prices)
+
+    benchmark_first_date = get_first_date(benchmark_prices)
+    benchmark_last_date = get_last_date(benchmark_prices)
```

```
+
@SaktiJaga ➜/workspaces/esilv-marketpulse-g07-totally_spies (feature/display-observation-dates) $ git commit -m "feat:
display first and last observation dates"
[feature/display-observation-dates 0723915] feat: display first and last observation dates
 1 file changed, 24 insertions(+), 6 deletions(-)
@SaktiJaga ➜/workspaces/esilv-marketpulse-g07-totally_spies (feature/display-observation-dates) $ git status
On branch feature/display-observation-dates
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
@SaktiJaga ➜/workspaces/esilv-marketpulse-g07-totally_spies (feature/display-observation-dates) $ git log --oneline -5
0723915 (HEAD -> feature/display-observation-dates) feat: display first and last observation dates
c388ccd (origin/main, origin/HEAD) Add files via upload
5207db8 first td session
ba5c9f4 Update Readme
9fd6c8a Update Readme
@SaktiJaga ➜/workspaces/esilv-marketpulse-g07-totally_spies (feature/display-observation-dates) $
```

Part 6 - Compare feature branch and main

```
@SaktiJaga ➜/workspaces/esilv-marketpulse-g07-totally_spies (feature/display-observation-dates) $ git log --oneline --g
raph --decorate --all
| * 1e32a1a (BrancheSakti) docs: add header comment to main entry point
| | *   30729f9 (origin/yanis_branche) Merge branch 'plensy:main' into yanis_branche
| | |\
| |_|/
|/| |
| | * 4bf5366 TD01
| | | * 63d1303 (origin/BrancheSakti) Merge branch 'plensy:main' into BrancheSakti
| | |/|
| |/|/
| |/|
|/| |
| * | 1aacee8 TD1&2
| |/
| | *   e5c2437 (origin/romybranche) Merge branch 'plensy:main' into romybranche
| | |\
| |_|/
|/| |
* | | c388ccd (origin/main, origin/HEAD) Add files via upload
|/ /
| * 53e1e6b TD1-2
|/
| * 9f95abb (origin/BrancheFatima) td1&2
|/
| * 93ae071 (origin/emmabranche) TD1 ET 2
|/
* 5207db8 first td session
```

```
@SaktiJaga ➜/workspaces/esilv-marketpulse-g07-totally_spies (feature/display-observation-dates) $ git diff origin/main..
.feature/display-observation-dates
diff --git a/src/main.py b/src/main.py
index 884b045..1b342e5 100644
--- a/src/main.py
+++ b/src/main.py
@@ -25,6 +25,12 @@ def load_prices():
 def filter_prices(prices, ticker):
     return [row for row in prices if row["ticker"] == ticker]

+def get_first_date(prices):
+    return prices[0]["date"]
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+
+def get_last_date(prices):
+    return prices[-1]["date"]

 def main():
     instruments = load_instruments()
@@ -39,22 +45,34 @@ def main():
     instrument_latest = instrument_prices[-1]
     benchmark_latest = benchmark_prices[-1]

+    instrument_first_date = get_first_date(instrument_prices)
+    instrument_last_date = get_last_date(instrument_prices)
+
+    benchmark_first_date = get_first_date(benchmark_prices)
+    benchmark_last_date = get_last_date(benchmark_prices)
```

```
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (feature/display-observation-dates) $ git status
On branch feature/display-observation-dates
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

Part 7 - Merge the feature into main

```
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (feature/display-observation-dates) $ git switch main
branch 'main' set up to track 'origin/main'.
Switched to a new branch 'main'
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (main) $ git branch
  BrancheSakti
  feature/display-observation-dates
* main
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (main) $ git merge feature/display-observation-dates
Updating c388ccd..0723915
Fast-forward
 src/main.py | 30 ++++++++++++++++++++++------
 1 file changed, 24 insertions(+), 6 deletions(-)
```

```
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (main) $ git log --oneline --graph --decorate --all
* 0723915 (HEAD -> main, feature/display-observation-dates) feat: display first and last observation dates
| * 1e32a1a (BrancheSakti) docs: add header comment to main entry point
| | *   30729f9 (origin/yanis_branche) Merge branch 'plensy:main' into yanis_branche
| | |\
| |_|/
|/| |
| | * 4bf5366 TD01
| | | * 63d1303 (origin/BrancheSakti) Merge branch 'plensy:main' into BrancheSakti
| | |/|
| |/|/
| |/|
|/| |
| * | 1aacee8 TD1&2
| |/
| | *   e5c2437 (origin/romybranche) Merge branch 'plensy:main' into romybranche
| | |\
| |_|/
|/| |
* | | c388ccd (origin/main, origin/HEAD) Add files via upload
|/ /
| * 53e1e6b TD1-2
|/
| * 9f95abb (origin/BrancheFatima) td1&2
|/
| * 93ae071 (origin/emmabranche) TD1 ET 2
|/
* 5207db8 first td session
```

```
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (main) $ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Observations : 21
First date   : 2026-09-01
Last date    : 2026-09-30
First close  : 250.00 USD
Last close   : 266.20 USD

Benchmark
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First close  : 6600.00
Last close   : 6742.00

Period: 1 month
Interval: Daily
```

1. Is the feature now visible from main?

Yes. The feature has been successfully merged into the main branch. The changes are now accessible from main.

2. Is the feature commit still visible?

Yes. The feature commit 0723915 remains visible in the Git history. Both main and the feature branch point to this commit.

3. Did Git perform a fast-forward or create a merge commit?

Git performed a fast-forward merge. Since main had no new diverging commits, Git simply advanced its pointer to the feature commit without creating an additional merge commit.

4. What does the graph show?

The graph shows that main and feature/display-observation-dates now point to the same commit. The feature has been integrated into main, while the feature branch still exists.

The remote branch origin/main has not yet been updated because the changes have not been pushed to the remote repository.

Part 8 - Capture Checkpoint A branch evidence, then clean the branch

```
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (main) $ git branch
  BrancheSakti
  feature/display-observation-dates
* main
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (main) $ git branch --merged
  feature/display-observation-dates
* main
```

```
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (main) $ git log --oneline --graph --decorate --all
* 0723915 (HEAD -> main, feature/display-observation-dates) feat: display first and last observation dates
| * 1e32a1a (BrancheSakti) docs: add header comment to main entry point
| | *   30729f9 (origin/yanis_branche) Merge branch 'plensy:main' into yanis_branche
| | |\
| |_|/
|/| |
| | * 4bf5366 TD01
| | | * 63d1303 (origin/BrancheSakti) Merge branch 'plensy:main' into BrancheSakti
| | |/|
| |/|/
| |/|
|/| |
| * | 1aacee8 TD1&2
| |/
| | *   e5c2437 (origin/romybranche) Merge branch 'plensy:main' into romybranche
| | |\
| |_|/
|/| |
* | | c388ccd (origin/main, origin/HEAD) Add files via upload
|/ /
| * 53e1e6b TD1-2
|/
| * 9f95abb (origin/BrancheFatima) td1&2
|/
| * 93ae071 (origin/emmabranche) TD1 ET 2
|/
* 5207db8 first td session
```